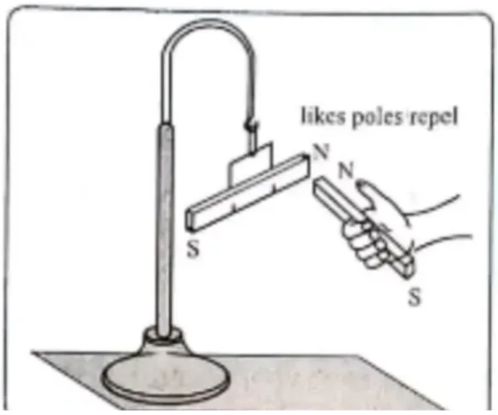
EXPERIMENT NO. 12
μ_1 / μ_2 BY SUSPENSION METHOD

Aim: Comparison of magnetic dipole moments (m) of a bar magnet by Individual, and by Assist and Oppose Method.

Apparatus: A pairs of bar magnets (A & B) (preferably having different magnetic moment), retort stand, thread, stop watch, etc.

Diagram :



$(m_1 + m_2)$ (b)
(Assist Position)

$(m_1 - m_2)$ (c)
(Oppose position)

Formula :

$$\mu = 4\pi^2 \frac{I}{B_H T^2}$$

Observations :

1. Least count of Vernier callipers = 0.01 cm.
2. Zero error of Vernier calipers = Z = ...0......cm
3. Mass of the Bar Magnet (A) = ----45---- g.
4. Mass of the Bar Magnet (B) = ----30---- g.

Observation Table for measurement of length and breadth

Obs no.	Magnet	Position	M.S.R Cm a	V.S.D.	V.S.R. cm b	T.R. cm c=a+b	Corrected reading d=c-z cm	Mean cm
1	A	Length	7·7	3	0·03	7·73	7·73	L_1= 2·73
			7·7	3	0·03	7·73	7·73	
2	A	Breadth	1·2	9	0·09	1·29	1·29	B_1= 1·21
			1·1	3	0·01	1·13	1·13	
3	B	Length	5·1	1	0·01	5·11	5·11	L_2= 5·11
			5·1	2	0·02	5·12	5·12	
4	B	Breadth	1·1	1	0·01	1·11	1·11	B_2= 1·14
			1·1	8	0·08	1·18	1·18	

Observation table for periodic time (T)

Ob. No.	Magnet	Time for 20 oscillations t second			Time period T =t/20 second
		t_1	t_2	Mean t	
1	1 Individual	29	30	29·5	T_1= 2·95
2	2 individual	23	22	22·5	T_2= 2·25
3	Assist (1+2)	35	34	34·5	T_S= 3·45
4	Oppose (1-2)	56	57	56·5	T_D= 5·65

Calculations:

Moment of Inertia of Magnet (A)

$$I_1 = M\left[\frac{L_1^2 + B_1^2}{12}\right] = 45\left[\frac{(7\cdot73)^2 + (1\cdot21)^2}{12}\right]$$

$$= 45 \times 3\cdot10 = 229$$

Moment of Inertia of Magnet (B)

$$I_2 = M\left[\frac{L_2^2 + B_2^2}{12}\right] = 30\left[\frac{(5\cdot11)^2 + (1\cdot14)^2}{12}\right]$$

$$= 30 \times 2\cdot3 = 68\cdot5$$

Individual Method

$$\frac{\mu_1}{\mu_2} = \frac{I_1}{I_2} \times \left(\frac{T_1}{T_2}\right)^2 = \frac{229\cdot5}{68\cdot5} \times \left[\frac{2\cdot25}{2\cdot95}\right]^2$$

Result :

1. The magnetic moments of the magnets were compared using Individual, Sum and difference method
2. $\frac{\mu_1}{\mu_2}$ = 1:9.43. (**Individual Method**)
3. $\frac{\mu_1}{\mu_2}$ = 2:1.8 (**Assist and oppose Method**)

Questions

1.What is the principle of a suspension magnetometer?

This device works on the principle that whenever a freely suspended magnet in a uniform magnetic field is distrbed from it's equilibrium position it starts vibrating about the mean position. In shorts we can say vibration magnetometer work on the principle or torque acting on the bar magnet.

2.Can you calculate magnetic dipole moment of the bar magnets? How?

A magnetic dipole consists of two equal & opposite magnetic charges having pole strength +m and -m seperated by finite distance.

Magnetic dipole moment the magnetic dipole moment is defined as the product of the pole strength and the magnetic length of a product.

3.Why do we take small oscillations while measuring time period?

Allows a short response time however, small oscillations, lead no the identification of irrelevant minima & maxima.